

Computational Optics, Kinematic Engineering, and Photochemical Inversion in Christopher Nolan's *Tenet*

Source guide

This technical analysis explores the complex cinematographic engineering required to realize Christopher Nolan's vision for *Tenet* using a strictly photochemical workflow. To visualize the film's "time inversion" without digital shortcuts, the production utilized modified 65mm and 70mm cameras capable of running film in reverse while maintaining precise mechanical and optical stability. The text details the kinematic modeling of massive film transport systems and the creation of customized optics designed to handle extreme close-ups and unique lighting constraints. Ultimately, the source highlights how these analog innovations and rigorous previsualization methods preserved the highest possible image resolution, proving that large-format film remains a versatile and powerful medium for modern, high-action storytelling

The cinematography of Christopher Nolan's *Tenet* represents a highly sophisticated application of physical engineering, thermodynamic theories, and large-format photochemical science.[1, 2, 3] Directed by Nolan and lensed by cinematographer Hoyte van Hoytema, the production rejected standard digital intermediate post-production in favor of an entirely photochemical workflow, from the camera negative to the final theatrical print.[1, 4, 5] This aesthetic and technical commitment required the physical modification of motion picture cameras and optical systems to capture "time inversion" directly on film emulsion.[1, 5] By forcing physical systems to run backward and modifying optics to operate under extreme physical constraints, the filmmakers expanded the mechanical limits of 65mm and 70mm transport technologies.[1, 6, 7]

Kinematic Modeling of Film Transport and Aspect Ratio Geometry

The acquisition strategy of *Tenet* relied on a strategic mixture of 15-perforation horizontal 65mm/70mm IMAX and 5-perforation vertical 65mm Panavision and Arriflex systems.[1, 4, 7, 8] Because IMAX cameras operate with a high level of acoustic noise due to the massive physical forces required to transport the large-format film horizontally, they are structurally unsuited for recording dialogue.[4, 7, 8, 9] Consequently, scenes requiring audible speech were captured on quieter, vertical 5-perforation 65mm systems, such as the Panavision System 65 and the Arriflex 765.[1, 4, 7, 8]

To analyze the physical dynamics of these large-format systems, the linear velocity of the film transport must be modeled. Let V represent the linear velocity of the film stock (in meters per minute), F represent the operational frame rate (in frames per second), P represent the number of perforations per frame, and p

perf

represent the standard pitch spacing between perforations, standardized at 4.75 mm (0.00475 m) for KS-type perforations. The kinematic transport velocity is defined by:

$$V = F \cdot P \cdot p$$

perf

$$\cdot 60$$

For the horizontal 15-perforation IMAX format, the linear velocity calculation at the standard capture speed of 24 fps is:

$$V$$

IMAX

$$= 24 \cdot 15 \cdot 0.00475 \cdot 60 = 102.6 \text{ m/min} \approx 337 \text{ ft/min}$$

Conversely, the vertical 5-perforation 65mm format operates under the following parameters:

$$V$$

5-perf

$$= 24 \cdot 5 \cdot 0.00475 \cdot 60 = 34.2 \text{ m/min} \approx 112 \text{ ft/min}$$

This threefold difference in film velocity demonstrates why the mechanical stresses exerted on the film negative within the IMAX camera are so extreme.[10] Accelerating a 65mm wide strip of polyester base to over 102 m/min intermittently—stopping and starting the film 24 times every second—requires powerful drive motors and precise registration mechanics.[10, 11, 12]

Parameter	IMAX 15-Perf 65/70mm	Panavision System 65 / Arriflex 765 5-Perf
Film Transport Orientation	Horizontal [10, 13]	Vertical [10]

Aperture Dimensions	69.6 mm×48.5 mm [10]	48.5 mm×22.1 mm [10]
Perforations per Frame (<i>P</i>)	15 [10, 13]	5 [10]
Linear Film Velocity (<i>V</i>)	102.7 m/min (337 ft/min) [7, 10]	34.2 m/min (112 ft/min) [10]
Camera Weight (Standard Rig)	≈40 kg (88 lbs) [14]	≈15 to 32 kg (33 to 70 lbs) [15, 16]
Approximate Resolving Power	≈18K equivalent [17]	≈6K–8K equivalent [18]

These formats forced a sophisticated, multi-ratio framing strategy.[14] As explained by 1st AC Bob Hall, the crew had to compose shots with two distinct aspect ratios in mind for dialogue scenes (where they used System 65 or Arriflex 765) and three aspect ratios for all other sequences.[14] The primary compositional priority was the theatrical widescreen aspect ratio of 2.39:1 (often referred to as 2.40:1), while simultaneously protecting the vertically expanded 1.43:1 full-height frame for 15/70 IMAX projection.[14, 19] This geometric balance ensured that the composition remained tight for widescreen theatrical exhibition without losing headroom or exposing microphone booms in the vertically expanded IMAX frame.[14]

This spatial constraint is further complicated by the sheer mass of the physical film prints.[13] A single 1-hour reel of 15/70 IMAX film weighs approximately 300 lbs (136 kg), while a full 3-hour theatrical print tips the scales at a staggering 600 lbs (272 kg).[13, 20] Managing these prints requires specialized platter systems and projection architectures capable of pulling the film through the loop at a speed of 337 ft/min.[13]

The Dynamics of Photochemical Inversion and Mechanical Reverse Transport

The narrative foundation of *Tenet* centers on the concept of entropy reversal, where inverted characters and objects move backward through time while the surrounding world moves forward.[1, 2, 3] To capture these scenes photochemically, the filmmakers decided to shoot action sequences twice: once with the actors performing their movements forward, and once with

them performing their choreography in reverse.[1, 18] Crucially, to maintain the correct motion cadence and organic highlight roll-off of the film stock, the cameras themselves had to physically run the film negative backward.[5, 21, 22]

Reversing a standard motion picture camera is an intricate mechanical challenge.[5] While standard cameras are designed to transport film in a single direction, Nolan's reliance on a photochemical finish required the engineering of specialized "reverse" magazines for both the 15-perforation IMAX cameras (MSM 9802, MKIII, and MKIV) and the 5-perforation Panavision 65mm cameras.[5, 22]

Standard Forward Transport:

—> —> (Slipping Clutch)

Reverse Transport Modification:

<— <—

In a standard magazine, a slipping clutch assembly on the take-up spindle regulates tension as the physical radius of the film roll $r(t)$ increases, ensuring the linear tension T remains constant [15]:

$$T =$$

$$r(t)$$

$$\tau$$

Where τ is the torque exerted by the spindle motor.[15] When running the camera in reverse, the original supply spindle must actively act as the tension-regulated take-up spindle, while the original take-up spindle must provide passive, controlled hold-back tension.[5, 15] The custom-built reverse magazines incorporated modified internal motor clutches and bi-directional tensioning assemblies to handle this shift.[5, 15, 22]

To prevent registration errors, the pulldown claw mechanisms in the Arriflex 765 and IMAX cameras were calibrated to ensure steady frame placement regardless of transport direction.[15] The Arriflex 765 achieves this steadiness through a patented multi-link movement with a dual 3-tooth pulldown claw and dual single-pin registration system, keeping frame placement steady to within 0.1% of the frame height (approximately 0.0009 inches).[15] To minimize noise and vibration during dialogue scenes, the Arriflex 765 separates its drive system into four

independent crystal-sync motors—two in the camera body and two in the magazine—reducing acoustic noise levels to below 25 dBA.[15]

To manage these complex magazine configurations on set, the camera crew utilized a strict, color-coded marking system [23]:

- **Red Tape:** Identified standard forward-running magazines.[23]
- **Blue Tape:** Identified the reverse-engineered magazines.[23]

Additionally, the cameras were retrofitted with high-definition video assist taps integrated with Blackmagic Design playback units.[5] This configuration enabled the crew to capture the analog gate image, instantly reverse the digital video feed, and play it back in real-time.[5] This step was vital to confirm that the complex choreography of forward and reverse actors aligned precisely within the frame.[5, 18]

Previsualization and Spatial Alignment of Dual-Temporal Physics

The interaction between forward-moving characters and inverted characters creates a causality paradox.[18, 24] Coordinating these complex "temporal pincer movements" required a rigorous blocking methodology.[18] DNEG VFX Supervisor Andrew Jackson used Autodesk Maya to build detailed 3D previsualizations, notably for the multi-car highway chase featuring the Saab 9-5.[18, 24, 25, 26]

To ensure the safety of the stunt performers and maintain spatial coherence, the trajectories of forward and backward actors had to be mapped simultaneously.[18] Mathematically, the spatial coordinates of a forward character $A(t)$ and an inverted character $B(T-t)$ must intersect at exact spatial-temporal coordinates (x,y,z,t) . [18] By blocking these timelines in Maya, the animators and camera crew could scrub forward and backward through the timeline, validating that visual cues, vehicle placements, and debris patterns remained structurally consistent from both forward and backward camera perspectives.[18]

This previsualization process also helped the stunt department physically prepare for the shoot.[18] Performers filmed their actions, reversed the footage digitally, and studied the reversed clips to learn how the weight and balance of their bodies would translate when the footage was eventually reversed in the final cut.[18] This virtual training allowed the stunt department and camera operators to rehearse physical actions—such as running and fighting in reverse—while understanding how their movements would look in both directions.[18]

Computational Exposure and Shutter Mechanics under Variable Frequency

Maintaining consistent exposure and motion blur across various international shooting locations required careful calibration of the camera's mechanical shutter.[1, 27] Film cameras like the Arriflex 765 utilize an adjustable rotating mirror reflex shutter to control exposure time.[15] The exposure duration t_e

t_e

is mathematically defined as a function of the frame rate F and the shutter angle θ [27]:

t_e

t_e

=

360

◦

· F

θ

At the standard theatrical frame rate of 24 fps and a standard 180

◦

shutter angle, the exposure time is:

t_e

t_e

=

360

◦

· 24

180

◦

=

48

1

seconds

When filming in locations with 50 Hz alternating current (AC) electrical grids, such as the UK and Europe, shooting at 24 fps with a standard 180

◦

shutter can introduce exposure flicker from magnetic ballast lighting.[27] This flicker occurs because the AC line frequency cycles 100 times per second (two half-cycles per wave), which is out of sync with a 1/48 second exposure time.[27] To prevent this, the shutter angle is adjusted to 172.8

◦

to sync the exposure time with the utility frequency [15, 16]:

t

e

=

360

◦

· 24

172.8

◦

=

50

1

seconds

To maintain identical exposure when altering the mechanical shutter angle, the lens aperture must be adjusted to compensate for the change in light.[27] The exposure change in stops (Δstops) is calculated using the ratio of the initial shutter angle θ

1

to the new shutter angle θ

2

:

$\Delta\text{stops}=\log$

2

(

θ

2

θ

1

)

When dropping the shutter angle from a standard θ

1

=172.8

◦

to a narrow θ

2

=86.4

◦

to achieve a crisp, low-motion-blur aesthetic for action sequences, the exposure loss is exactly:

$\Delta\text{stops}=\log$

2

(

86.4

◦

172.8

◦

)=1 stop

To maintain consistent exposure on the film negative, the lens iris must be opened by exactly one full stop (for example, from $T4.0$ to $T2.8$).[27]

Alternating Current Frequency	Frame Rate (F)	Shutter Angle (θ)	Calculated Exposure Time (t_e)	Stop Compensation	Practical Application
60 Hz	24 fps	180.0 °	$\frac{48}{1}$ $s \approx 20.83 \text{ ms}$	Base (0 stops)	Standard US daylight capture [27]
60 Hz	24 fps	144.0 °	$\frac{60}{1}$ $s \approx 16.67 \text{ ms}$	−0.32 stops	Anti-flicker for 60 Hz areas [15]
50 Hz	24 fps	172.8 °	$\frac{50}{1}$ $s \approx 20.00 \text{ ms}$	−0.06 stops	Standard UK/EU capture [15, 16]
50 Hz	24 fps	86.4 °	$\frac{100}{1}$ $s \approx 10.00 \text{ ms}$	−1.06 stops	Narrow shutter action sequencing [27]

Optical Customization and Aberration Control by Dan Sasaki

The naturalistic style of *Tenet* relied heavily on available light and close-proximity framing.[4, 7, 8, 21] To achieve this look on large-format film, Hoyte van Hoytema collaborated closely with Dan Sasaki, Panavision’s Senior Vice President of Optical Engineering and Lens Strategy.[1, 7]

The massive dimensions of the 15-perforation 70mm frame create an extremely shallow depth of field.[10, 14] Converted Hasselblad medium-format lenses and Panavision Sphero 65 spherical

lenses were selected as the primary optics, avoiding the clinical look of modern aspheric lenses.[7, 19, 28] Sasaki customized these lenses to meet three specific criteria: boosting lens speed to a fast $T2.0$ to support low-light capture, modifying the focus mechanics to reduce minimum-focus limits down to 8 inches, and eliminating optical distortions like focus breathing.[7, 29]

[Hasselblad Medium-Format Glass]

|
▼ (Re-engineered by Dan Sasaki)

1. Optimized Internal Spherical Element Spacing [7]	---> Achieved T2.0 Aperture Speed
2. Modified Focus Helicoid Assembly [7, 29]	---> Reduced Minimum Focus to 8 Inches
3. Elimination of Non-Linear Edge Distortions [7]	---> Preserved 1:1 "Micro-Macro" Rendering

Rather than seeking mathematical perfection, Sasaki focused on crafting lenses with distinct "personalities" and gentle contrast roll-offs.[7, 30] To support close-up shooting on the massive IMAX format, Sasaki built specialized "micro-macro" 50mm and 80mm lenses.[7] These optics could achieve a 1:1 magnification ratio, allowing the camera to capture detailed close-ups completely free of "mumps" (anamorphic field distortions or non-linear focus breathing).[7, 29]

Lens Series	Focal Lengths Used	Max Aperture (T -Stop)	Close Focus Limit	Optical Character & Aberration Management
Panavision Sphero 65	80 mm (Primary Portrait) [7]	$T2.0$ [7]	Standard	Moderate focus roll-off, blended layers of contrast, and pleasant skin tones.[7]

Custom Hasselblad Spherical	50 mm, 80 mm [7, 28]	<i>T</i> 2.0 (Modified) [7]	8 inches (Modified) [7, 29]	Re-engineered helicoids for extreme close-focus; optimized for close-up portraits.[7, 29]
Hasselblad "Micro-Macro"	50 mm, 80 mm [7]	<i>T</i> 2.8 [7]	1:1 Magnification [7]	Optimized for zero geometric distortion at macro ranges; used for detailed scientific inserts.[7]

Kinematic Mobility and the Integration of the Logmar Magellan

One of the defining stylistic choices of modern large-format cinematography is the transition from heavy, tripod-mounted setups to dynamic, handheld cameras.[4, 14, 21] Because standard 15-perforation IMAX cameras weigh over 40 kg when fully loaded, operating them handheld for extended periods is physically grueling.[4, 14]

To address this challenge during the London shoot, the production deployed a prototype of the Danish Logmar Magellan 65mm camera.[1, 16] Developed by Tommy Lau Madsen and Lasse Røedtnes, the Magellan was the first newly designed 65mm motion picture camera built in three decades.[16, 31, 32, 33]

Traditional Large-Format Setup (IMAX MSM 9802):

- Physical Weight: >40 kg (88 lbs) [14]
- Optical Viewfinder: Rigid mechanical design limits low-angle operating [16, 34]
- Mobility Profile: Heavy support systems required (Tripods, Cranes, Car Mounts) [4]

Optimized Handheld Setup (Logmar Magellan):

- Physical Weight: <15 kg (33 lbs) [16]
- Viewfinder System: Flexible electronic viewfinder with real-time 3G-SDI tap [16, 34]
- Mobility Profile: Highly agile; suitable for shoulder-mounted, drone, and Steadicam use [16, 32]

The Logmar Magellan introduced several key innovations that improved mobility on set:

- **Lightweight Construction:** By utilizing modern alloys and carbon fiber components, the ready-to-shoot camera weighed under 15 kg (33 lbs), including the lens, battery, and a 500 ft film magazine.[16, 34] This lightweight design allowed the camera to be operated on Steadicams, compact remote heads, and specialized drones.[16, 32, 33]
- **Electronic Viewfinder and HD Tap:** Unlike traditional optical viewfinders, the Magellan integrated an electronic viewfinder and a native 3G-SDI 1080p60 tap.[16, 34] This configuration allowed the operators to frame shots comfortably in tight spaces and let the director monitor the performance in real-time.[16, 34]
- **Dual Registration Design:** The camera featured a custom dual-registration system that held the film securely at the center of the gate during exposure.[16] This mechanical design ensured maximum image sharpness while allowing the gate to be cleaned easily between takes.[16]

By incorporating the compact Logmar Magellan alongside their customized IMAX and Panavision cameras, the filmmakers were able to maintain a consistent large-format look while shooting dynamic, close-quarters action.[1, 34]

Photochemical Modulation Transfer Function and Contact Printing

Evaluating the spatial resolving power of the lens-and-film system requires modeling its spatial frequency response.[35] The Modulation Transfer Function (MTF) of the photochemical emulsion can be modeled using a Lorentzian approximation [35]:

$$MTF(f)=$$

$$1+($$

f

50

f

)

2

Where f is the spatial frequency in line pairs per millimeter (lp/mm) and f

50

is the spatial frequency where the response falls to 50%. [35]

For high-speed color negative films, the 50% response threshold is located at approximately f

50

=40 lp/mm. [35] To translate this spatial frequency into resolution relative to Picture Height (PH), we multiply the frequency by the vertical dimension of the film gate ($h=52.0$ mm for the IMAX negative area) [10, 35]:

MTF

IMAX,50

$=2 \cdot f$

50

$\cdot h = 2 \cdot 40 \cdot 52 = 4160$ Line Widths per Picture Height (LW/PH)

For the 20% response threshold, which represents the limit of fine detail resolution, the spatial frequency f

20

is derived from the Lorentzian model:

$0.20 =$

$1 + ($

f

50

f

20

)

2

1

$\Rightarrow f$

20

$=2 \cdot f$

50

$=80 \text{ lp/mm}$

This corresponds to a high-resolution detail limit of:

MTF

IMAX,20

$=2 \cdot f$

20

$\cdot h=2 \cdot 80 \cdot 52=8320 \text{ LW/PH}$

With high-contrast camera negatives, Kodak Vision3 stocks achieve resolving powers up to 72 cycles/mm (equivalent to 142 LW/mm).[7, 35] Because every step of photochemical

duplication introduces spatial frequency losses, the final theatrical print resolution is defined by the product of each generation's *MTF*:

MTF

system

$(f) = MTF$

lens

$(f) \cdot MTF$

negative

$(f) \cdot MTF$

interpositive

$(f) \cdot MTF$

internegative

$(f) \cdot MTF$

projector

(f)

This cascade of resolution loss highlights why Nolan's decision to bypass digital intermediates and create direct contact prints was so critical.[5, 17] By minimizing the generations of duplication, the production preserved the maximum resolving power of the original 15-perforation master negative.[17, 35]

To preserve this high-fidelity image during projection, theatrical presentation is handled by specialized 15/70 projection systems.[10, 11, 13] Unlike standard projectors that pull the film vertically using high-stress claws, the IMAX projector uses the "Rolling Loop" system

(originally designed by Peter Ronald Wright Jones and commercialized by Shaw).[10, 11] This system feeds horizontal waves of film into the gate using pressurized air jets, holding each frame against a field flattener lens via a vacuum suction system and registration pins to guarantee image stability on massive screens.[11]

By combining these engineering solutions, the filmmakers proved that high-resolution, large-format film could be used outside of static landscape photography, bringing a dynamic, handheld realism to complex action sequences.[4, 14, 21] The innovations developed for *Tenet* created a solid foundation for subsequent large-format productions, ensuring that analog cinematography remains a vital, evolving tool for modern visual storytelling.[21, 31, 33, 36]

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